

Related topics

Diffraction spectrometer, spin, angular momentum, spin-orbital angular momentum interaction, multiplicity, energy level, excitation energy, selection rules, doublets, parahelium, orthohelium, exchange energy, angular momentum, singlet series, triplet series, selection rules, forbidden transitions.

Principle

The well-known spectral lines of He are used for calibrating the diffraction spectrometer. The wave-lengths of the spectral lines of Na are determined using the spectrometer.

The prism spectrometer is calibrated with the aid of the He spectrum. The wavelengths of the spectral lines of Hg, Cd and Zn are determined.

Equipment

Spectrometer/goniom. w. vernier	35635.02	1
Diffraction grating, 600 lines/mm	08546.00	1
Spectral lamp He, pico 9 base	08120.03	1
Spectral lamp Na, pico 9 base	08120.07	1
Spectral lamp Hg 100, pico 9 base	08120.14	1
Spectral lamp Cd, pico 9 base	08120.01	1
Spectral lamp Zn, pico 9 base	08120.11	1
Power supply for spectral lamps	13662.97	1
Lamp holder, pico 9, f. spectr. lamps	08119.00	1
Tripod base -PASS-	02002.55	1

Tasks

1. Calibration of the spectrometer using the He spectrum, and the determination of the constant of the grating;

2. Determination of the spectrum of Na;
3. Determination of the fine structure splitting.
4. Determination of the most intense spectral lines of Hg, Cd and Zn.

Set-up and procedure

The experimental set up is as shown in Fig. 1. The spectrometer/goniometer and the grating must be set up and adjusted according to the operating instructions.

In the second-order spectrum, the sodium D-line is split. The micrometer screw is set to 0 and the cross hairs in the telescope positioned to coincide with the red line (2nd-order). The telescope is locked by means of the knurled head screw. The cross hairs are first positioned at the long-wave and then at the short-wave sodium D-line, with the micrometer screw, the particular micrometer positions being noted each time. It is also possible to measure the splitting starting from the shortwave side. The only essential is that the direction of rotation of the micrometer screw is maintained, otherwise the play in the micrometer spindle might lead to errors. When measuring in the reverse direction, the micrometer screw must be set to 10 and the cross hairs in the telescope again positioned to coincide with the red line (2nd-order). For quantitative determination of wavelengths, the micrometer screw must be calibrated round the entire circle. The spectral lamps attain their full illuminating power after being warmed up for about 5 minutes. The lamp housing should be adjusted so that air can circulate freely through the ventilation slits. Before changing the spectral lamps a cooling period must be allowed since the paper towels or cloths used in this operation might otherwise stick to the glass of the lamp.

Fig.1: Experimental set up for determining the spectral lines of Na.



Theory and evaluation

1. If light of a wavelength λ falls on a grating of constant d it is diffracted. Intensity maxima are produced if the angle of diffraction α which satisfies the following conditions:

$$n \cdot \lambda = d \cdot \sin \alpha; n = 0, 1, 2 \dots$$

red	667.8 nm
yellow	587.6 nm
green	501.6 nm
greenish blue	492.2 nm
bluish green	471.3 nm
blue	447.1 nm

Table 1: Wavelength of the He spectrum.

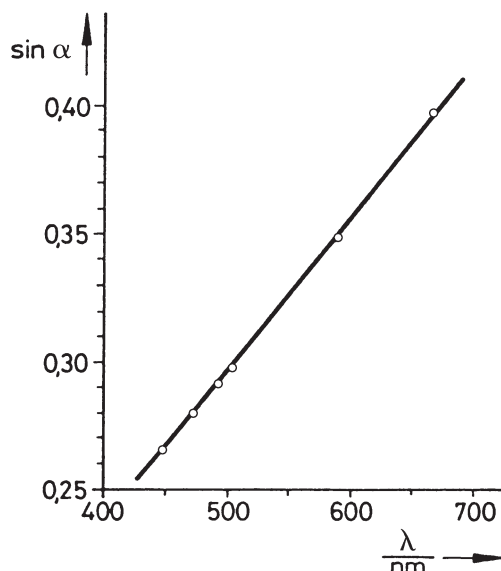


Fig. 2: Calibration curve of the diffraction spectrometer.

The calibration curve of the diffraction spectrometer (Fig. 2) is plotted for the first order ($n = 1$) and the measured angles α .

The grating constant is

$$d = 1684 \text{ nm.}$$

This value may vary for different gratings.

2. The excitation of the Na atoms is produced by electron impact. The energy difference produced by the return of electrons from the excited level E_1 to the original state E_0 is emitted as a photon, of frequency f , given by:

$$hf = E_1 - E_0$$

where

$$h = \text{Planck's constant} \\ = 6.63 \cdot 10^{-34} \text{ Js.}$$

To a first approximation the electrons of the inner complete shell produce a screening of the potential V due to the charge on the nucleus, as regards the single external electron, but the potential is position-dependent:

$$V(r) = - \frac{e^2 Z_{\text{eff}}(r)}{4 \pi \epsilon_0 r},$$

where e is the charge of the electron.

The energy levels are similar to those of hydrogen, with reduced degeneracy of angular momentum.

$$E_{nl} = - \frac{me^2}{8\hbar^2} z_{nl}^2 \frac{1}{n^2}$$

An approximation formula for E_{nl} is given below:

$$E_{nl} = - \frac{me^2}{8\hbar^2} \frac{1}{(n - \mu_{nl})^2} \quad (1)$$

The quantum defect μ_{nl} depends to some slight extent on n and decreases as l increases.

$n \setminus l$	0	1	2	3	4
3	1.35	0.85	0.01		
4				0.00	
5					0.00

Table 2: μ_{nl} of the Na atom.

The interaction of the spin $\vec{S}$ of the electron with its orbital moment gives rise to a reduction in the degeneracy of the total angular momentum:

$$j = \left| l + \frac{1}{2} \right| \dots \left| l - \frac{1}{2} \right|,$$

where l is the orbital angular momentum of the external electron.

If we consider the interaction term in perturbation theory:

$$H = \xi(r) \vec{S} \cdot \vec{l}$$

we obtain the following for (1).

$$E_{nlj} = E_{nl} + \xi_{nl} \frac{1}{2} (j(j+1) - S(S+1) - l(l+1))$$

and as splitting:

$$E_{nlj} = l + \frac{1}{2} - E_{nlj} = l - \frac{1}{2} (2l+1) \xi_{nl}.$$

The following lines of the Na atom were measured in the first order spectrum:

red	617.2 nm
yellow	588.4 nm
yellowish green	567.7 nm
green	514.1 nm
greenish blue	498.7 nm

Table 3: Experimentally determined Na wavelengths.

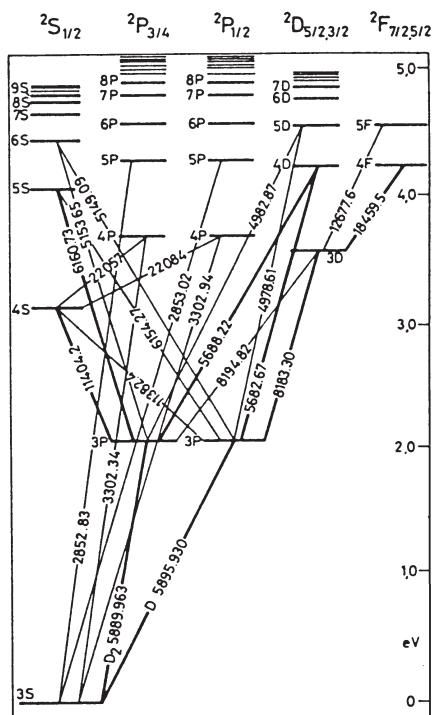


Fig. 3: Spectrum of sodium.

The separation of the yellow D-line was determined in the second-order spectrum. First of all, the wavelength of the shorter sodium D-line in the second order spectrum was determined:

$$\lambda_1 = 588.6 \text{ nm.}$$

The difference between the short-wave and the long-wave sodium D-line was then determined using the micrometer screw:

$$\lambda_2 - \lambda_1 = 0.614 \text{ nm.}$$

Set-up and procedure with prism spectrometer

The experimental set up is as shown in Fig. 4. The spectrometer/goniometer and the prism must be set up and adjusted in accordance with the operating instructions.

The spectral lamps attain their maximum light intensity after a warm-up period of approx. 5 min. The lamp housing should be set up so as to ensure free circulation of air through the ventilator slit. Before changing the spectral lamps they must be allowed to cool since the paper towels or cloths used for this operation might otherwise stick to the glass. The illuminated scale is used for recording the spectra.

Theory and evaluation

When light of wavelength λ passes through a prism, it is deviated. The angle of deviation depends on the geometry of the prism and on the angle of incidence. The refractive index of a prism depends on the wavelength and thus also on the angle of deviation. Fig. 5 shows the calibration curve for the He spectrum (dispersion curve), obtained at the angle of minimum deviation.

Excitation of atoms results from electron impact. The energy difference produced when electrons revert from the excited state E_0 is emitted as a photon with a frequency f .

$$hf = E_1 - E_0$$

where

$$h = \text{Planck's constant} \\ = 6.63 \cdot 10^{-34} \text{ Js.}$$

Fig. 4: Experimental set up for measuring the spectra of Hg, Cd and Zn.



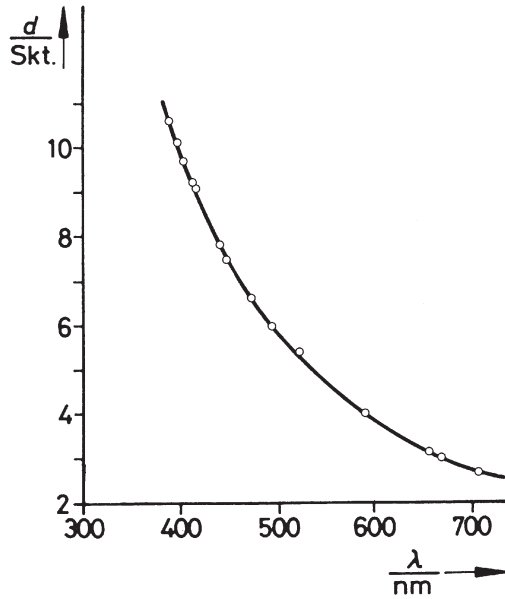


Fig. 5: Calibration curve of the prism spectrometer.

The Hamiltonian operator (non-relativistic) for the two electrons 1 and 2 of the He atom is:

$$H = -\frac{\hbar^2}{2m}\Delta_1 - \frac{\hbar^2}{2m}\Delta_2 - \frac{2e^2}{|\vec{r}_1|} - \frac{2e^2}{|\vec{r}_2|} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$$

where $\hbar = \frac{h}{2\pi}$,

m and e represent the mass and charge of the electron respectively,

$$\Delta_i = \frac{d^2}{dx_i^2} + \frac{d^2}{dy_i^2} + \frac{d^2}{dz_i^2}$$

is the Laplace operator, and $\vec{r}_i$ is the position of the i -th electron. The Spin-orbit interaction energy

$$E_{so} \propto \frac{Z^4}{4 \cdot (137)^2}$$

was ignored in the case of the nuclear charge $Z = 2$ of helium, because it is small when Z is small.

If we consider

$$\frac{e}{|\vec{r}_1 - \vec{r}_2|}$$

as the electron-electron interaction term, then the eigenvalues of the Hamiltonian operator without interaction are those of the hydrogen atom:

$$E_{n,m}^0 = -\frac{me^2}{8h^2} \left(\frac{1}{n^2} + \frac{1}{m^2} \right)$$

$$n, m = 1, 2, 3, \dots$$

As the transition probability for simultaneous two-electron excitation is very much less than that for one-electron excitation, the energy spectrum of the undisturbed system is:

$$E_{n,m}^0 = -\frac{me^2}{8h^2} \left(1 + \frac{1}{m^2} \right)$$

$$m = 1, 2$$

The interaction term removes the angular momentum degeneracy of the pure hydrogen spectrum and the exchange energy degeneracy. There results an energy adjustment:

$$E_{nl\pm}^1 = \langle \phi_{nl\alpha}^\pm | \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} | \phi_{nl\alpha}^\pm \rangle = C_{nl} \pm A_{nl}$$

in which $\phi_{nl\alpha}^\pm$ are the antisymmetrized undisturbed 2-particle wave functions with symmetrical (ϕ^+) or antisymmetrical (ϕ^-) position component, l^* is the angular momentum quantum number, and α is the set of the other quantum numbers required.

In the present case, the orbital angular momentum of the single electron l is equal to the total angular momentum of the two electrons L , since only one-particle excitations are being considered and the second electron remains in the ground state ($l = 0$).

C_{nl} and A_{nl} are the Coulomb and exchange energy respectively. They are positive. Coupling the orbital angular momentum L with the total spin S produces for $S = 0$, i.e. ϕ^+ , a singlet series and for $S = 1$, i.e. ϕ^- , a triplet series. Because of the lack of spin-orbit interaction, splitting within a triplet is slight. As the disturbed wave functions are eigenfunctions for S^2 and as S^2 interchanges with the dipole operator, the selection rule

$$\Delta S = 0$$

(which is characteristic for 2-electron systems with a low nuclear charge number) results and forbids transitions between the triplet and singlet levels.

In addition, independent of the spin-orbit interaction, the selection rule for the total angular momentum

$$\Delta J = 0, \pm 1$$

applies except where

$$J = 0 \rightarrow J' = 0$$

If the spin-orbit interaction is slight, then

$$\Delta L = 0, \pm 1$$

applies.

Detailed calculations produce the helium spectrum of Fig. 6.

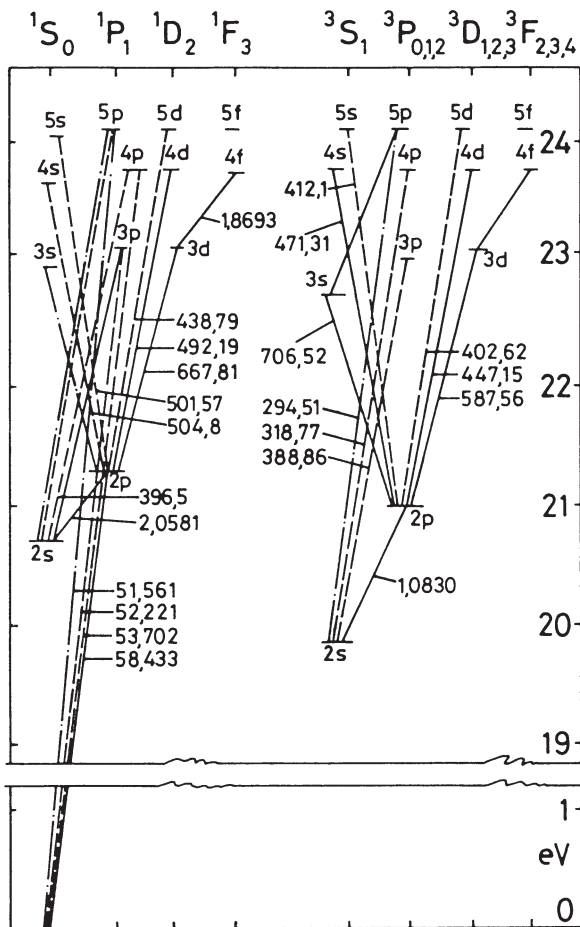


Fig. 6: Spectrum of helium.

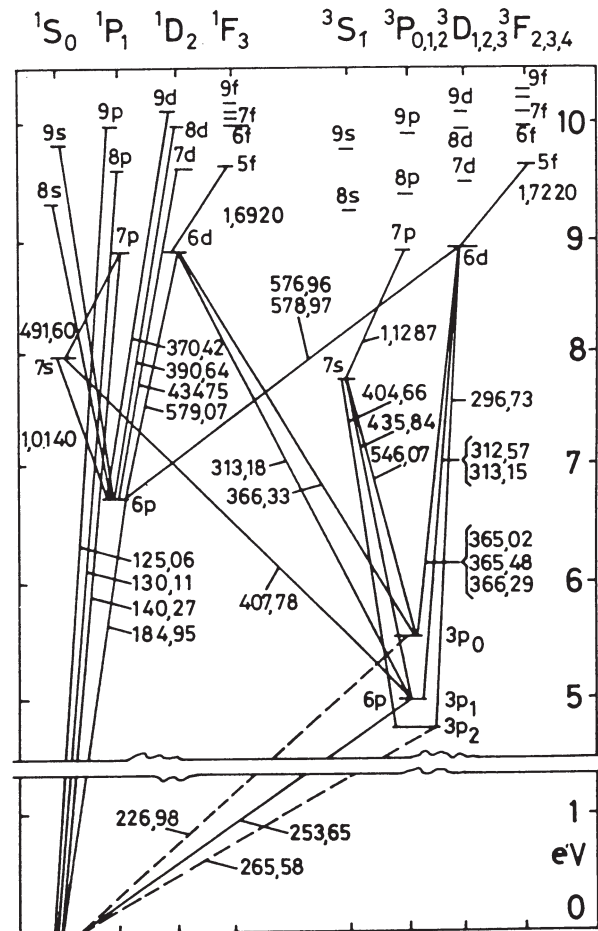


Fig. 7: Spectrum of mercury.

Hg, Cd and Zn are also two-electron systems and possess the structure of 2 series.

The spin-orbit interaction, however, is relatively pronounced so that only the total angular momentum

$$J = L + S$$

Colour	λ/nm	Transition	Relative intensity
red	706.5	$3^3S \rightarrow 2^1P$	5
red	667.8	$3^1D \rightarrow 2^1P$	6
red	656.0	He II	4-6
yellow	587.6	$3^3D \rightarrow 2^3P$	10
green	504.8	$4^1S \rightarrow 2^1P$	2
green	492.2	$4^1D \rightarrow 2^1P$	4
blue	471.3	$4^3S \rightarrow 2^3P$	3
blue	447.1	$4^3D \rightarrow 2^3P$	6
blue	438.8	$5^1D \rightarrow 2^1P$	3
violet	414.4	$6^1D \rightarrow 2^1P$	2
violet	412.1	$5^3S \rightarrow 2^3P$	3
violet	402.6	$5^3D \rightarrow 2^3P$	5
violet	396.5	$4^1P \rightarrow 2^1S$	4
violet	388.9	$3^3P \rightarrow 2^3S$	10

Table 4: He-I spectrum.

is an energy conservation parameter. Splitting within a triplet is pronounced. Moreover, the selection rule

$$\Delta S = 0$$

is no longer valid since S is no longer a conservation parameter (transition from $L-S$ for the $j-j$ coupling).

Colour	λ/nm	Transition
red	690	$8^3P_2 \rightarrow 7^3S$
red	624	$9^1P_1 \rightarrow 7^1S$
red	611	$8^1P_1 \rightarrow 7^3S$
red	608	$8^1P_1 \rightarrow 7^1S$
yellow	578	$\left\{ \begin{array}{l} 6^3D_2, 6^3D_1 \\ 6^1D_2 \rightarrow 6^1P_1 \end{array} \right.$
green	548	$7^3S \rightarrow 6^3P_1$
blue-green	496	Hg II
blue-green	492	$8^1D \rightarrow 6^1P_1$
blue	435	$7^1D \rightarrow 6^1P$
violet	408	$7^1S \rightarrow 6^3P_1$

Table 5: Measured Hg-1 spectrum

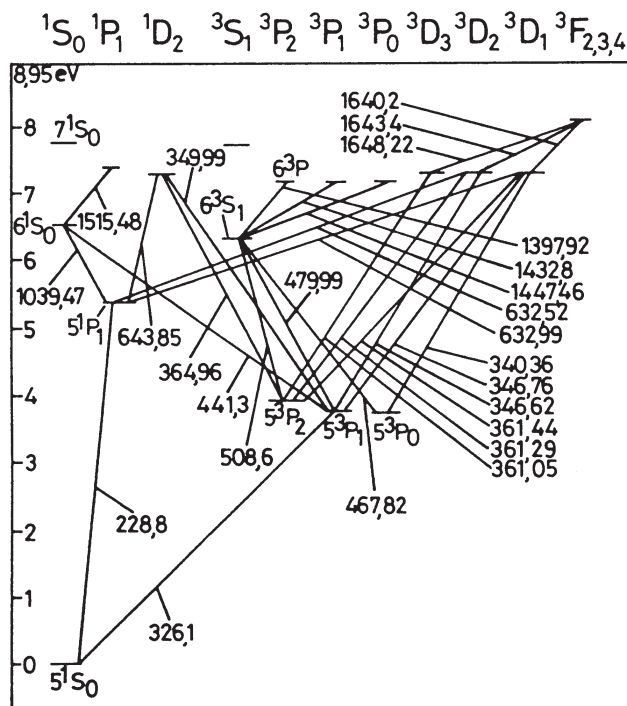


Fig. 8: Spectrum of Cd.

Colour	λ/nm	Transition
red	645	$6^1D_2 \rightarrow 5^1P_1$
red	633	$5^3D_1 \rightarrow 5^1P_1$
green	517	$7^1S_0 \rightarrow 5^1P_1$
green	509	$6^3S_1 \rightarrow 5^3P_2$
blue	480	$6^3S_1 \rightarrow 5^3P_1$
blue	469	$6^3S_1 \rightarrow 5^3P_0$
violet	441	$6^1S_0 \rightarrow 5^3P_1$

Table 6: Measured Cd spectrum.

Colour	λ/nm	Transition
red	636	$4^1P_1 \rightarrow 4^1D_1$
yellow	589	ZN II
yellow	579	$5^3S_1 \rightarrow 7^3P_2$
		$5^3S_1 \rightarrow 7^3P_1$
		$5^3S_1 \rightarrow 8^3P_0$
green	534	$5^3S_1 \rightarrow 9^3P_1$
green	519	$4^1P_1 \rightarrow 6^1S_0$
green	508	$5^3S_1 \rightarrow 9^3P_1$
blue	481	$4^3P_2 \rightarrow 5^3S_1$
blue	472	$4^3P_1 \rightarrow 5^3S_1$
blue	468	$4^3P_0 \rightarrow 5^3S_1$
violet	463	$4^1P_1 \rightarrow f^1D_2$
violet	429	$4^3P_1 \rightarrow 5^1S_0$
		$4^1P_1 \rightarrow 7^1S_0$

Table 7: Measured Zn spectrum.